Efficiency Loss of Sequential Quasi-Maximum Likelihood Estimation in the AR(p)-EGARCH(1,1) Model

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Abstract

This paper compares joint and sequential Gaussian quasi-maximum likelihood estimation of the AR(p)-EGARCH(1,1) model. The two-step (2S) estimator fits the mean equation by ordinary least squares and then the variance equation by quasi-maximum likelihood, and the three-step (3S) estimator refits the mean equation by feasible generalized least squares. For symmetric innovations with finite exponential moments, we derive the information matrices and, under high-level asymptotic expansions, the asymptotic distributions of the three estimators. Block diagonality of the information matrix fails even without a leverage effect, through a single entry that links the mean return to the coefficient on the lagged innovation in the log variance. With a leverage effect, the 2S estimator also loses efficiency for the intercept and the persistence of the log variance. The loss is largest for the unconditional mean of the log variance, approximately 15% in the base design of the Monte Carlo study, and 17% to 21% at the S&P 500 estimate. The 3S estimator recovers most of the efficiency for the mean parameters.[1]

1 Introduction

The generalized autoregressive conditional heteroskedasticity (GARCH) model of Bollerslev (1986) was extended by Nelson (1991) to the exponential GARCH (EGARCH) model, which "remains one of the most popular GARCH type models for modelling the volatility of financial time series" (Hafner and Linton, 2017). It allows for asymmetric effects of positive and negative shocks on volatility and guarantees a positive conditional variance by construction. In applications, the variance equation is combined with a model for the conditional mean, typically an autoregression of order p, AR(p), which accounts for the autocorrelation that discontinuous trading induces in index returns (Nelson, 1991). Nelson (1991) and Tsay (2010) estimate the two parts jointly, whereas Engle and Ng (1993) and Hafner and Linton (2017) fit the variance model to the residuals of a first-step mean regression.

Sequential estimation of conditionally heteroskedastic regression models was proposed by Engle (1982). In the regression model with ARCH errors and Gaussian innovations, the information matrix is block diagonal between the mean and the variance parameters under the symmetry and regularity conditions of Engle (1982, Theorem 4). The two sets of parameters can then be estimated separately without asymptotic loss of efficiency. Block diagonality extends to autoregressive moving average (ARMA) models with GARCH errors and a symmetric innovation, as established by Ling and McAleer (2003) and Francq and Zakoïan (2004, Theorem 3.2). Bollerslev et al. (1994) note

[1] Code and replication data are available at *GitHub*.

that it typically fails for EGARCH, in which the conditional variance is an asymmetric function of lagged residuals, but do not identify the nonzero off-diagonal entries or quantify the resulting efficiency loss.

We compare the joint Gaussian quasi-maximum likelihood estimator (QMLE), the J estimator, with two sequential estimators. The two-step (2S) estimator fits the mean by ordinary least squares (OLS) and then the variance parameters by Gaussian QMLE given the OLS residuals. The three-step (3S) estimator then refits the mean by generalized least squares (GLS) with the fitted variances, weighting each observation by the inverse of its fitted variance. For the AR(p)-EGARCH(1,1) model with a symmetric innovation that has finite exponential moments, we investigate which parameters the sequential estimators estimate less efficiently than the J estimator.

The comparison is conducted in terms of the asymptotic covariance matrices of the three estimators. Straumann and Mikosch (2006, Theorem 5.1) establish the strong consistency of the QMLE and Wintenberger (2013, Theorem 6) the asymptotic normality of the stable QMLE for the pure EGARCH(1,1) model, and Hafner and Kyriakopoulou (2021) establish asymptotic normality for the EGARCH-in-mean model. For an autoregression with asymmetric GARCH errors, whose variance depends on the absolute residual, Meitz and Saikkonen (2011) establish only consistency. Following Nelson (1991), who assumed consistency and asymptotic normality, we impose the asymptotic expansions of the estimators as a high-level assumption and derive the covariance matrices from them.

We prove that block diagonality fails even without leverage, when the log variance is symmetric in the lagged innovation. The off-diagonal block between the mean and the variance parameters then contains a single nonzero entry, which links the mean return to the sign effect, the coefficient on the lagged innovation in the log variance. Consequently, the 2S estimator loses efficiency only for the sign effect, and only when the mean return is estimated. This loss is strictly positive whenever the innovation kurtosis falls below an explicit threshold. With leverage, the entries linking the mean return to the remaining variance parameters are of first order in the sign effect, and the 2S loss in these parameters is of second order when the mean return is estimated. Numerical evaluation illustrates that the loss increases with persistence and, as an application to daily S&P 500 returns confirms, is largest for the unconditional mean of the log variance, approximately 15% in the base design of the Monte Carlo study and 17% to 21% at the S&P 500 estimate. The 3S estimator attains the efficiency of GLS with known variances, and without leverage the J estimator is at least as efficient as the 3S estimator for the mean parameters if and only if the innovation kurtosis does not exceed an explicit threshold.

The remainder of this paper is organized as follows. Section 2 states the model, the estimators and the assumptions. Section 3 derives the structure of the information matrices, Section 4 the asymptotic distributions under high-level expansions and Section 5 the efficiency comparisons. Sections 6 and 7 present the Monte Carlo study and the application to S&P 500 returns, and Section 8 concludes. The proofs are collected in Appendix A.

2 Model, estimators and assumptions

2.1 Model and quasi-likelihood

Let (y_t) be the observed returns. The AR(p)-EGARCH(1,1) model is

$$\begin{aligned} y_t &= \nu_0 + \sum_{i=1}^{p} \phi_{0i}(y_{t-i} - \nu_0) + \varepsilon_t, \qquad \varepsilon_t = \sigma_t \eta_t, \\ h_t &= \log \sigma_t^2 = \omega_0 + \alpha_0\big(|\eta_{t-1}| - \mathsf{E}|\eta_{t-1}|\big) + \gamma_0 \eta_{t-1} + \beta_0 h_{t-1}, \end{aligned} \tag{2.1}$$

where (η_t) is the innovation sequence, and the size term is centered at $\mathsf{E}|\eta_{t-1}|$ (Nelson, 1991). The size effect α_0 governs the log-variance response to innovation size, and the sign effect γ_0 governs the response to innovation sign (Hafner and Linton, 2017).

The parameter is $\boldsymbol{\theta} = [\boldsymbol{m}^\mathsf{T}, \boldsymbol{v}^\mathsf{T}]^\mathsf{T}$, with mean part $\boldsymbol{m} = [\nu, \boldsymbol{\phi}^\mathsf{T}]^\mathsf{T}$, $\boldsymbol{\phi} = [\phi_1, \ldots, \phi_p]^\mathsf{T}$, and variance part $\boldsymbol{v} = [\omega, \alpha, \gamma, \beta]^\mathsf{T}$. The true parameter is $\boldsymbol{\theta}_0 = [\nu_0, \boldsymbol{\phi}_0^\mathsf{T}, \omega_0, \alpha_0, \gamma_0, \beta_0]^\mathsf{T}$. Vectors and matrices are partitioned conformably with $\boldsymbol{\theta} = [\boldsymbol{m}^\mathsf{T}, \boldsymbol{v}^\mathsf{T}]^\mathsf{T}$. At a parameter $\boldsymbol{\theta}$, the residual and the filtered log variance are

$$\begin{aligned} \varepsilon_t(\boldsymbol{m}) &= y_t - \nu - \sum_{i=1}^{p} \phi_i(y_{t-i} - \nu), \\ h_t(\boldsymbol{\theta}) &= \omega + \alpha\big(|\varepsilon_{t-1}(\boldsymbol{m})| e^{-h_{t-1}(\boldsymbol{\theta})/2} - c\big) + \gamma\, \varepsilon_{t-1}(\boldsymbol{m}) e^{-h_{t-1}(\boldsymbol{\theta})/2} + \beta h_{t-1}(\boldsymbol{\theta}), \end{aligned} \tag{2.2}$$

where c is a known constant. Then $\varepsilon_t = \varepsilon_t(\boldsymbol{m}_0)$, and in Sections 2 to 6, $c = \mathsf{E}|\eta_t|$, so that h_t satisfies (2.2) at $\boldsymbol{\theta}_0$. Let $\mathcal{F}_t$ be the σ-field generated by $(\eta_s)_{s \le t}$. The Gaussian log quasi-likelihood of observation t is

$$l_t(\boldsymbol{\theta}) = -\frac{1}{2}\Big\{ h_t(\boldsymbol{\theta}) + \varepsilon_t(\boldsymbol{m})^2 e^{-h_t(\boldsymbol{\theta})} \Big\}. \tag{2.3}$$

Let $\Theta = \Theta_{\boldsymbol{m}} \times \Theta_{\boldsymbol{v}}$ be a compact subset of $\mathbb{R}^{p+1} \times \{\boldsymbol{v} : \beta \in [0,1),\ \alpha \ge |\gamma|\}$. The recursion (2.2) is started at a fixed value $\hat{h}_0 \ge \max_{\boldsymbol{\theta} \in \Theta} \underline{h}(\boldsymbol{\theta})$, where $\underline{h}(\boldsymbol{\theta}) = (\omega - \alpha c)/(1 - \beta)$, and $\hat{h}_t(\boldsymbol{\theta})$ and $\hat{l}_t(\boldsymbol{\theta})$ denote the resulting feasible quantities.

Definition 2.1 (Estimators)**.**

(i) The J estimator is

$$\hat{\boldsymbol{\theta}}_{\mathrm{J}} = \arg\max_{\boldsymbol{\theta} \in \Theta} \sum_{t=1}^{T} \hat{l}_t(\boldsymbol{\theta}).$$

(ii) The 2S estimator is $\hat{\boldsymbol{\theta}}_{\mathrm{2S}} = [\hat{\boldsymbol{m}}_1^\mathsf{T}, \hat{\boldsymbol{v}}_2^\mathsf{T}]^\mathsf{T}$. Its first step is the OLS estimator $\hat{\boldsymbol{m}}_1$ of the autoregression, and its second step is the Gaussian QMLE of the variance parameters given $\hat{\boldsymbol{m}}_1$,

$$\hat{\boldsymbol{v}}_2 = \arg\max_{\boldsymbol{v} \in \Theta_{\boldsymbol{v}}} \sum_{t} \hat{l}_t(\hat{\boldsymbol{m}}_1, \boldsymbol{v}).$$

(iii) The 3S estimator is $\hat{\boldsymbol{\theta}}_{\mathrm{3S}} = [\hat{\boldsymbol{m}}_3^\mathsf{T}, \hat{\boldsymbol{v}}_2^\mathsf{T}]^\mathsf{T}$. Its first two steps are those of the 2S estimator, and

its third step is the feasible GLS estimator of the mean parameters,

$$\hat{\boldsymbol{m}}_3 = \arg\min_{\boldsymbol{m}\in\Theta_{\boldsymbol{m}}} \sum_t \varepsilon_t(\boldsymbol{m})^2 e^{-\hat{h}_t(\hat{\boldsymbol{\theta}}_{2S})}.$$

The 2S and 3S estimators follow Engle (1982), except for the third step. Engle's third step is a scoring step for the regression parameters, which yields efficient estimates based only on a consistent estimate of the variance parameters. A one-step estimator performs a single Newton–Raphson iteration on the full parameter from a $\sqrt{T}$-consistent start, with a consistent estimate of the expected Hessian. Under the conditions of Newey and McFadden (1994, Theorem 3.5), it has the asymptotic variance of the J estimator with or without block diagonality, and Engle's scoring step has this property only under block diagonality. The losses derived below are those of the 2S and 3S estimators, which stop after the second or third step.

2.2 Assumptions

The filter (2.2) is a stochastic recurrence equation $h_t(\boldsymbol{\theta}) = \mathcal{T}_{t-1}^{\boldsymbol{\theta}}(h_{t-1}(\boldsymbol{\theta}))$ (Straumann and Mikosch, 2006) with random maps

$$\mathcal{T}_s^{\boldsymbol{\theta}}(x) = \omega + \alpha\big(|\varepsilon_s(\boldsymbol{m})|e^{-x/2} - c\big) + \gamma\,\varepsilon_s(\boldsymbol{m})e^{-x/2} + \beta x, \tag{2.4}$$

and we let

$$A_t = \beta_0 - \tfrac{1}{2}\big(\alpha_0|\eta_t| + \gamma_0\eta_t\big), \qquad \rho = \mathsf{E}[A_t^2]. \tag{2.5}$$

We impose the following assumptions.

(A1) (η_t) is independent and identically distributed (IID) with unit variance, and its distribution is continuous and symmetric about zero.

(A2) $\mathsf{E}[e^{\chi|\eta_t|}] < \infty$ for every $\chi > 0$.

(A3) $\beta_0 \in [0,1)$, $\alpha_0 > 0$, $\alpha_0 \geq |\gamma_0|$, $\rho < 1$, and the roots of the autoregressive polynomial lie outside the unit circle.

Remark 2.1. (A2) holds for the Gaussian distribution and for the generalized error distribution (GED) with shape parameter above one, and fails for every Student t distribution. It guarantees the moments of all orders of σ_t, ε_t and y_t required in the proofs.

Under (A1) to (A3), the model has a unique strictly stationary and ergodic solution in which σ_t and σ_t^{-1} have finite moments of all orders (Lemma A.1), and $\kappa_4 = \mathsf{E}[\eta_t^4]$ is finite. Differentiating (2.2) at $\boldsymbol{\theta}_0$ yields a linear stochastic recurrence equation for the filter derivative $\partial h_t/\partial\boldsymbol{\theta}$, which has a stationary solution in L^2 (Lemma A.2).

(A4) There is a closed ball $\mathcal{N} \subset \Theta$ around $\boldsymbol{\theta}_0$ such that the following hold.

(i) For every $\boldsymbol{\theta} \in \mathcal{N}$, the recursion (2.2) has a unique strictly stationary solution $h_t(\boldsymbol{\theta})$, which is $\mathcal{F}_{t-1}$-measurable.

(ii) $|h_t(\boldsymbol{\theta}) - h_t(\boldsymbol{\theta}')| \leq L_t|\boldsymbol{\theta} - \boldsymbol{\theta}'|$ for all $\boldsymbol{\theta}, \boldsymbol{\theta}' \in \mathcal{N}$ almost surely (a.s.), where $\mathsf{E}[L_t^q] < \infty$ for some $q > 2$.

(iii) For every $\boldsymbol{\theta} \in \mathcal{N}$, h_t is a.s. differentiable at $\boldsymbol{\theta}$, and $\mathsf{E}|\partial h_t(\boldsymbol{\theta})/\partial\boldsymbol{\theta} - \partial h_t(\boldsymbol{\theta}_0)/\partial\boldsymbol{\theta}|^2 \to 0$ as $\boldsymbol{\theta} \to \boldsymbol{\theta}_0$.

Remark 2.2. At $\boldsymbol{\theta}_0$, the filter is invertible if, from any starting value, $\hat{h}_t - h_t \to 0$ in probability, where h_t is the stationary solution (Straumann and Mikosch, 2006). Let $\Lambda_s^{(r)}(\boldsymbol{\theta})$ be the Lipschitz coefficient of $\mathcal{T}_s^{\boldsymbol{\theta}} \circ \cdots \circ \mathcal{T}_{s-r+1}^{\boldsymbol{\theta}}$ on $[\underline{h}(\boldsymbol{\theta}), \infty)$, an interval that each map (2.4) leaves invariant (Wintenberger, 2013) under the constraints $\beta \geq 0$ and $\alpha \geq |\gamma|$ of Straumann and Mikosch (2006, Proposition 3.9). The contraction condition INV($\boldsymbol{\theta}$) of Wintenberger (2013) requires $\mathsf{E}[\log \Lambda_s^{(r)}(\boldsymbol{\theta})] < 0$ for some $r \geq 1$, with logarithmic moments that follow from (A2). Under INV($\boldsymbol{\theta}_0$), Theorem 3.1 of Bougerol (1993) implies that the filter is invertible, and, with $\Lambda_s^{(r)}(\boldsymbol{\theta})$ continuous in $\boldsymbol{\theta}$, the proof of Theorem 2 of Wintenberger (2013) extends invertibility to a neighborhood of $\boldsymbol{\theta}_0$, on which Theorem 2.8 of Straumann and Mikosch (2006) yields (A4)(i).

3 Block structure of the information matrix

At $\boldsymbol{\theta}_0$, the score of (2.3) is

$$\boldsymbol{s}_t = -\frac{1}{2}(1 - \eta_t^2)\frac{\partial h_t}{\partial \boldsymbol{\theta}} - \eta_t\, \sigma_t^{-1}\frac{\partial \varepsilon_t}{\partial \boldsymbol{\theta}}, \tag{3.1}$$

where only the first term involves the variance parameters.

Proposition 3.1 (Information matrices)**.** *Under (A1) to (A4),*

$$\mathcal{I} = \mathsf{E}[\boldsymbol{s}_t\boldsymbol{s}_t^{\mathsf{T}}] = \frac{\kappa_4 - 1}{4}\, C + \operatorname{diag}(W, 0), \qquad \mathcal{J} = -\frac{\partial \mathsf{E}[\boldsymbol{s}_t(\boldsymbol{\theta})]}{\partial \boldsymbol{\theta}^{\mathsf{T}}}\Big|_{\boldsymbol{\theta}_0} = \frac{1}{2}\, C + \operatorname{diag}(W, 0), \tag{3.2}$$

where

$$C = \mathsf{E}\Big[\frac{\partial h_t}{\partial \boldsymbol{\theta}}\frac{\partial h_t}{\partial \boldsymbol{\theta}^{\mathsf{T}}}\Big], \qquad W = \mathsf{E}\Big[\sigma_t^{-2}\frac{\partial \varepsilon_t}{\partial \boldsymbol{m}}\frac{\partial \varepsilon_t}{\partial \boldsymbol{m}^{\mathsf{T}}}\Big]. \tag{3.3}$$

The matrices W and $\mathcal{J}$ are positive definite.

The matrix W is the information about $\boldsymbol{m}$ of the infeasible GLS regression with σ_t known. The J estimator additionally exploits the variance equation, so that

$$\mathcal{J}_{\boldsymbol{m}\boldsymbol{m}} = W + \tfrac{1}{2}C_{\boldsymbol{m}\boldsymbol{m}}, \qquad \mathcal{I}_{\boldsymbol{m}\boldsymbol{m}} = W + \tfrac{\kappa_4 - 1}{4}C_{\boldsymbol{m}\boldsymbol{m}}.$$

The matrix $\mathcal{J}$ is defined through the expected score, which is differentiable at $\boldsymbol{\theta}_0$, although the scores are not differentiable on the surfaces $\{\boldsymbol{\theta} : \varepsilon_s(\boldsymbol{m}) = 0\}$ (Newey and McFadden, 1994, Theorem 7.3). By (3.2), $\mathcal{J}_{\boldsymbol{m}\boldsymbol{v}} = \frac{1}{2}C_{\boldsymbol{m}\boldsymbol{v}}$ and $\mathcal{I}_{\boldsymbol{m}\boldsymbol{v}} = \frac{\kappa_4 - 1}{4}C_{\boldsymbol{m}\boldsymbol{v}}$, so both matrices are block diagonal between $\boldsymbol{m}$ and $\boldsymbol{v}$ if and only if $C_{\boldsymbol{m}\boldsymbol{v}} = 0$.

Theorem 3.2 (Block structure without leverage)**.** *Under (A1) to (A3) with $\gamma_0 = 0$, partition $\boldsymbol{\theta}$ into $\{\boldsymbol{\phi}\}$, $\{\nu, \gamma\}$ and $\{\omega, \alpha, \beta\}$.*

(i) C and W are block diagonal with respect to this partition, and so are $\mathcal{I}$ and $\mathcal{J}$ under (A4).

(ii) $C_{\nu\gamma}$ and $C_{\gamma\gamma}$ are

$$C_{\nu\gamma} = -\frac{\alpha_0\,\Phi(1)\,\mathsf{E}|\eta_t|\,\mathsf{E}[\sigma_t^{-1}]}{1-\rho}, \qquad C_{\gamma\gamma} = \frac{1}{1-\rho}, \tag{3.4}$$

where $\Phi(1) = 1 - \sum_i \phi_{0i}$, so that $C_{\nu\gamma} \neq 0$.

(iii) With

$$\tau = \frac{\alpha_0^2}{1-\rho}, \qquad \varpi = \frac{(\mathsf{E}|\eta_t|)^2(\mathsf{E}[\sigma_t^{-1}])^2}{\mathsf{E}[\sigma_t^{-2}]} \in (0,1), \tag{3.5}$$

we have $C_{\boldsymbol{mm}} = \tau W$ and $C_{\nu\nu} - C_{\nu\gamma}^2/C_{\gamma\gamma} = \tau(1-\varpi)W_{\nu\nu}$.

Proposition 3.3 (Off-diagonal entries under leverage)**.** *Let (A1) to (A3) hold at $\gamma_0 = 0$, and let $C(\gamma_0)$ denote C as a function of γ_0, with the other parameters and the distribution of η_t held fixed. Then $C(\gamma_0)$ is real analytic near $\gamma_0 = 0$, and, with $\bar{\boldsymbol{v}} = (\omega, \alpha, \beta)$,*

$$C_{\boldsymbol{\phi}\gamma} = O(\gamma_0), \qquad C_{(\nu,\gamma)\bar{\boldsymbol{v}}} = O(\gamma_0), \qquad C_{\boldsymbol{\phi}\bar{\boldsymbol{v}}} = O(\gamma_0^2). \tag{3.6}$$

4 Asymptotic distributions

Write $\boldsymbol{\xi}_t = \partial\varepsilon_t/\partial\boldsymbol{m}$, $\boldsymbol{b}_t^{(\boldsymbol{m})} = \sigma_t^{-1}\boldsymbol{\xi}_t$ and $M = \mathsf{E}[\boldsymbol{\xi}_t\boldsymbol{\xi}_t^{\mathsf{T}}]$. By Lemma A.3, the OLS estimator satisfies

$$\sqrt{T}(\hat{\boldsymbol{m}}_1 - \boldsymbol{m}_0) = T^{-1/2}\sum_t \boldsymbol{p}_t + o_p(1), \qquad \boldsymbol{p}_t = -M^{-1}\boldsymbol{\xi}_t\varepsilon_t, \tag{4.1}$$

and has the asymptotic covariance

$$V_{\mathrm{OLS}} = M^{-1}\mathsf{E}\big[\sigma_t^2\boldsymbol{\xi}_t\boldsymbol{\xi}_t^{\mathsf{T}}\big]M^{-1}. \tag{4.2}$$

The asymptotic analysis is based on the following high-level assumption, which postulates expansions of the form that standard regularity conditions imply for the J estimator (Bollerslev and Wooldridge, 1992) and for sequential estimators (Newey, 1984; Murphy and Topel, 1985).

(A5) The J, 2S and 3S estimators satisfy

$$\sqrt{T}(\hat{\boldsymbol{\theta}}_{\mathrm{J}} - \boldsymbol{\theta}_0) = \mathcal{J}^{-1}\,T^{-1/2}\sum_{t=1}^{T}\boldsymbol{s}_t + o_p(1), \tag{4.3}$$

$$\sqrt{T}(\hat{\boldsymbol{v}}_2 - \boldsymbol{v}_0) = \mathcal{J}_{\boldsymbol{vv}}^{-1}\,T^{-1/2}\sum_{t=1}^{T}\big(\boldsymbol{s}_t^{(\boldsymbol{v})} - \mathcal{J}_{\boldsymbol{vm}}\boldsymbol{p}_t\big) + o_p(1), \tag{4.4}$$

$$\sqrt{T}(\hat{\boldsymbol{m}}_3 - \boldsymbol{m}_0) = -W^{-1}\,T^{-1/2}\sum_{t=1}^{T}\eta_t\boldsymbol{b}_t^{(\boldsymbol{m})} + o_p(1). \tag{4.5}$$

The 3S expansion (4.5) has no first-step correction. Since (ε_t) is a martingale difference sequence and $\boldsymbol{\xi}_t$ and $h_t(\boldsymbol{\theta})$ are $\mathcal{F}_{t-1}$-measurable, $\mathsf{E}\big[\varepsilon_t \boldsymbol{\xi}_t e^{-h_t(\boldsymbol{\theta})}\big] = \mathbf{0}$ for $\boldsymbol{\theta} \in \mathcal{N}$, so the derivative with respect to the first-step parameters vanishes at $\boldsymbol{\theta}_0$ (Newey and McFadden, 1994, Theorem 6.1).

Proposition 4.1 (Asymptotic normality)**.** *Under (A1) to (A5),*

$$\sqrt{T}(\hat{\boldsymbol{v}}_2 - \boldsymbol{v}_0) \xrightarrow{\mathrm{d}} \mathrm{N}(\mathbf{0}, V_{2\mathrm{S},\boldsymbol{vv}}), \qquad \sqrt{T}(\hat{\boldsymbol{m}}_3 - \boldsymbol{m}_0) \xrightarrow{\mathrm{d}} \mathrm{N}(\mathbf{0}, W^{-1}), \qquad \sqrt{T}(\hat{\boldsymbol{\theta}}_{\mathrm{J}} - \boldsymbol{\theta}_0) \xrightarrow{\mathrm{d}} \mathrm{N}(\mathbf{0}, V_{\mathrm{J}}),$$

where

$$V_{\mathrm{J}} = \mathcal{J}^{-1}\mathcal{I}\mathcal{J}^{-1}, \qquad V_{2\mathrm{S},\boldsymbol{vv}} = \mathcal{J}_{\boldsymbol{vv}}^{-1}\big(\mathcal{I}_{\boldsymbol{vv}} + \mathcal{J}_{\boldsymbol{vm}} V_{\mathrm{OLS}} \mathcal{J}_{\boldsymbol{mv}}\big)\mathcal{J}_{\boldsymbol{vv}}^{-1}. \tag{4.6}$$

Remark 4.1. The correction $\mathcal{J}_{\boldsymbol{vm}} V_{\mathrm{OLS}} \mathcal{J}_{\boldsymbol{mv}}$ in (4.6) is the first-step term of the two-step covariance of Newey (1984), and the cross terms of this covariance vanish because the OLS score and the variance score are uncorrelated under a symmetric innovation (Lemma A.4).

5 Relative efficiency of the sequential estimators

5.1 2S estimation of the variance parameters

Theorem 5.1 (2S versus J)**.** *Under (A1) to (A5),*

$$V_{2\mathrm{S},\boldsymbol{vv}} - V_{\mathrm{J},\boldsymbol{vv}} = \mathcal{J}_{\boldsymbol{vv}}^{-1}\mathcal{J}_{\boldsymbol{vm}}\big(V_{\mathrm{OLS}} - V_{\mathrm{J},\boldsymbol{mm}}\big)\mathcal{J}_{\boldsymbol{mv}}\mathcal{J}_{\boldsymbol{vv}}^{-1}. \tag{5.1}$$

Remark 5.1. The two estimators of $\boldsymbol{v}$ differ only in the estimate of $\boldsymbol{m}$ on which they condition, OLS for the 2S estimator and the J estimate for the J estimator. The loss is the product of the off-diagonal block $\mathcal{J}_{\boldsymbol{vm}} = \frac{1}{2}C_{\boldsymbol{vm}}$ and the efficiency gap $V_{\mathrm{OLS}} - V_{\mathrm{J},\boldsymbol{mm}}$ between the two mean estimates. The condition $\mathcal{J}_{\boldsymbol{vm}} = 0$ is sufficient for equality, and necessary when the gap is positive definite.

Let

$$\mathcal{R}(\kappa, x) = \frac{1 + (\kappa - 1)x/4}{(1 + x/2)^2} = 1 + \frac{x(\kappa - 5 - x)}{4(1 + x/2)^2}, \tag{5.2}$$

so that, for $x > 0$, $\mathcal{R}(\kappa, x) \leq 1$ if and only if $\kappa \leq 5 + x$.

Corollary 5.2 (No leverage)**.** *Under the assumptions of Theorem 5.1 with $\gamma_0 = 0$, $V_{2\mathrm{S},\boldsymbol{vv}}$ and $V_{\mathrm{J},\boldsymbol{vv}}$ can differ only in the (γ,γ) entry, and*

$$V_{2\mathrm{S},\gamma\gamma} - V_{\mathrm{J},\gamma\gamma} = \left(\frac{C_{\nu\gamma}}{C_{\gamma\gamma}}\right)^2 \big(V_{\mathrm{OLS},\nu\nu} - V_{\mathrm{J},\nu\nu}\big), \qquad V_{\mathrm{J},\nu\nu} = \frac{\mathcal{R}(\kappa_4, \tau(1-\varpi))}{W_{\nu\nu}}. \tag{5.3}$$

The difference is strictly positive if $\kappa_4 < 5 + \tau(1-\varpi)$. If ν is known, with parameter $[\boldsymbol{\phi}^{\mathsf{T}}, \boldsymbol{v}^{\mathsf{T}}]^{\mathsf{T}}$ and OLS without an intercept, and (A4) and (A5) hold for this parameter, $V_{2\mathrm{S},\boldsymbol{vv}} = V_{\mathrm{J},\boldsymbol{vv}}$.

Corollary 5.3 (Small leverage)**.** *Let ν be known (Corollary 5.2) and let the assumptions of Theorem 5.1, with (A4) and (A5) for $[\boldsymbol{\phi}^{\mathsf{T}}, \boldsymbol{v}^{\mathsf{T}}]^{\mathsf{T}}$, hold for every γ_0 near zero. With the asymptotic relative efficiency of the J to the 2S estimator, $\mathsf{ARE}_v = V_{2\mathrm{S},vv}/V_{\mathrm{J},vv}$,*

$$\mathsf{ARE}_\gamma(\gamma_0) - 1 = O(\gamma_0^2), \qquad \mathsf{ARE}_v(\gamma_0) - 1 = O(\gamma_0^4) \quad (v = \omega, \alpha, \beta). \tag{5.4}$$

Remark 5.2. With ν estimated, the 2S loss in ω, α and β is of order γ_0^2, against γ_0^4 with ν known (Corollary 5.3). By (3.6), the $\bar{\boldsymbol{v}}$ rows of $\mathcal{J}_{\boldsymbol{vv}}^{-1}\mathcal{J}_{\boldsymbol{vm}}$ in (5.1) are $O(\gamma_0)$ in the ν column, where $C_{\gamma\nu} \neq 0$ by Theorem 3.2(ii), and $O(\gamma_0^2)$ in the $\boldsymbol{\phi}$ columns, so the quadratic form (5.1) is of order γ_0^2.

Remark 5.3. Demeaning the returns does not avoid the loss from estimating ν. The sample mean has the influence function $\Phi(1)^{-1}\varepsilon_t$ of the OLS estimate of ν, so the first step has the influence function $\boldsymbol{p}_t$ with or without demeaning, and the variance parameters fitted to demeaned returns (Hafner and Linton, 2017) have the asymptotic covariance $V_{2\mathrm{S},\boldsymbol{vv}}$ of Proposition 4.1.

5.2 3S estimation of the mean parameters

The gap $V_{\mathrm{OLS}} - V_{\mathrm{J},\boldsymbol{mm}}$ in Theorem 5.1 splits at W^{-1},

$$V_{\mathrm{OLS}} - V_{\mathrm{J},\boldsymbol{mm}} = \left(V_{\mathrm{OLS}} - W^{-1}\right) + \left(W^{-1} - V_{\mathrm{J},\boldsymbol{mm}}\right), \tag{5.5}$$

into the gain from weighting by the conditional variance and the difference due to exploiting the variance equation. By Proposition 4.1, the 3S estimator attains W^{-1} for every κ_4.

Proposition 5.4 (Gain from weighting)**.** *Under (A1) to (A3), $V_{\mathrm{OLS}} - W^{-1}$ is positive semidefinite.*

Corollary 5.5 (Kurtosis thresholds)**.** *Under (A1) to (A5) with $\gamma_0 = 0$,*

$$V_{\mathrm{J},\boldsymbol{\phi\phi}} = \mathcal{R}(\kappa_4, \tau)\, W_{\boldsymbol{\phi\phi}}^{-1}, \tag{5.6}$$

and $V_{\mathrm{J},\nu\nu}$ is given in (5.3). Hence, the J estimator is at least as efficient as the 3S estimator for $\boldsymbol{\phi}$ if and only if $\kappa_4 \leq 5 + \tau$, and for ν if and only if $\kappa_4 \leq 5 + \tau(1 - \varpi)$.

6 Monte Carlo study

We assess the finite-sample performance of the OLS, 2S, 3S and J estimators and the accuracy of the population variance ratios. We generate $R = 1000$ replications of sample size $T \in \{1000, 10000\}$ from the AR(1)-EGARCH(1,1) model with $c = \mathsf{E}|\eta_t|$, after a burn-in of 1000 observations, and compute the estimates with ν estimated and with ν known.[2] Design 1, the base design, sets

$$\boldsymbol{\theta}_0 = [\nu_0, \phi_0, \omega_0, \alpha_0, \gamma_0, \beta_0]^{\mathsf{T}} = [0,\ 0.05,\ 0,\ 0.15,\ -0.08,\ 0.98]^{\mathsf{T}}.$$

The value $\omega_0 = 0$ makes the ratio for ω equal to that for the mean log variance $\omega/(1-\beta)$, which does not depend on c. Designs 2 to 5 each change one factor that governs the loss in Section 5. Design

[2] The filter starts at the logarithm of the mean squared OLS residual, which lies in the invariant interval of Remark 2.2 on a neighborhood of the estimates. The quasi-likelihoods are maximized by the BFGS algorithm with analytic gradients and relative tolerance 10^{-12}, in the parameterization $\alpha = e^a$, $\gamma = \alpha \tanh g$, $\beta = 1/(1 + e^{-b})$ and $\phi = \tanh f$, starting the variance parameters at $[0.1\hat{h}_0, 0.1, 0, 0.9]^{\mathsf{T}}$ and the J estimator at the 2S estimate. The algorithm restarts from its last point after 1000 iterations. At $T = 1000$, it fails to converge in 41, 0, 104, 18 and 35 samples of Designs 1 to 5, nearly all with an estimate on the boundary $|\gamma| = \alpha$ of Θ, and these samples are discarded. At $T = 10000$, all optimizations converge. Population values are computed from a path of 2×10^7 observations of the recursion for $\partial h_t/\partial\boldsymbol{\theta}$, except for ν in Design 2, where they follow from the closed forms of Theorem 3.2 and Corollary 5.2.

2 sets $\gamma_0 = 0$, the case of Corollary 5.2. Design 3 sets $\gamma_0 = -0.10$, the J estimate of Section 7, to examine the order γ_0^2 of Remark 5.2. Design 4 sets $\beta_0 = 0.95$, which shortens the half-life of a log-variance shock. Designs 1 to 4 use Gaussian innovations, while Design 5 draws η_t from the GED with shape parameter 1.5 and $\kappa_4 = 3.76$. In every design, $\mathsf{E}[\log \Lambda_s^{(20)}(\boldsymbol{\theta}_0)]$ lies between -1.03 and -0.31, so INV($\boldsymbol{\theta}_0$) holds with $r = 20$.

		$T = 1000$			$T = 10000$		
		Bias	SD	AD	Bias	SD	AD
ν	OLS	0.0020	0.0370	0.0366	0.0003	0.0118	0.0116
	3S	−0.0032	0.0318	0.0306	0.0000	0.0099	0.0097
	J	−0.0031	0.0313	0.0301	0.0000	0.0097	0.0095
ϕ	OLS	−0.0011	0.0407	0.0415	−0.0006	0.0132	0.0131
	3S	−0.0012	0.0341	0.0334	−0.0002	0.0108	0.0106
	J	−0.0013	0.0331	0.0321	0.0001	0.0103	0.0102
ω	2S	−0.0003	0.0055	0.0046	0.0000	0.0016	0.0015
	J	0.0001	0.0053	0.0043	0.0000	0.0015	0.0014
α	2S	−0.0041	0.0324	0.0306	−0.0003	0.0097	0.0097
	J	−0.0034	0.0325	0.0306	−0.0002	0.0097	0.0097
γ	2S	−0.0004	0.0193	0.0190	−0.0004	0.0063	0.0060
	J	−0.0013	0.0193	0.0187	−0.0005	0.0063	0.0059
β	2S	−0.0035	0.0098	0.0073	−0.0004	0.0022	0.0023
	J	−0.0035	0.0097	0.0073	−0.0004	0.0022	0.0023

Table 1: Bias, Monte Carlo standard deviation (SD) and asymptotic standard deviation (AD) of the estimates, Design 1 with ν estimated, over the converged samples of $R = 1000$. AD is $\sqrt{V_{ii}/T}$, with V the population covariance of Section 4. The OLS and 2S estimators share the mean estimates, and the 2S and 3S estimators the variance estimates.

Table 1 reports the bias, SD and AD of the estimates in Design 1. Every estimator has a small bias, and at $T = 10000$ its SD lies within 8% of its AD, which supports the normal approximation of Proposition 4.1. At $T = 1000$, the SD exceeds the AD by up to 22% for ω and 34% for β. The SD of OLS exceeds those of the 3S and J estimators by 16% to 28%.

Table 2 reports the variance ratios of the sequential estimators to the J estimator. With ν estimated, the 2S loss concentrates in ω, with population ratios of 1.148 in Design 1 and 1.250 in Design 3, whereas with ν known the ratio for ω is at most 1.002 in every design, consistent with Corollary 5.3. From Design 1 to Design 3, the loss in ω grows by a factor of 1.69, close to the factor 1.56 implied by Remark 5.2, and in Design 2 it vanishes, leaving a loss of 1.1% in γ (Corollary 5.2). Lower persistence reduces the ratio for ω to 1.036 in Design 4, whereas heavier tails leave it at 1.130 in Design 5. The losses in γ and β are at most 4.2% and 2.9%, and the ratio for α equals one (Table B.1). For the mean parameters, OLS has 1.18 to 1.90 times the variance of the J estimator, and the 3S estimator closes most of this gap, with ratios of 1.021 to 1.085. The J estimator remains the most efficient for the mean parameters in every design, and, in Design 2, $\kappa_4 = 3$ lies below the thresholds of Corollary 5.5, 5.15 for ϕ and 5.06 for ν. At $T = 10000$, every simulated ratio lies within three bootstrap standard errors of its population value, while at $T = 1000$ the simulated ratio for ω falls below it by up to 0.11, in Design 3.

	2S, γ		2S, ω		3S		OLS	
	ν estimated	ν known	ν estimated	ν known	ν	ϕ	ν	ϕ
Design 1								
Population	1.028	1.008	1.148	1.001	1.031	1.081	1.48	1.67
$T = 1000$	0.997	0.997	1.108	1.000	1.034	1.063	1.40	1.52
	(0.010)	(0.006)	(0.021)	(0.002)	(0.012)	(0.018)	(0.050)	(0.060)
$T = 10000$	1.016	1.007	1.142	1.001	1.027	1.089	1.47	1.64
	(0.010)	(0.006)	(0.026)	(0.002)	(0.011)	(0.018)	(0.053)	(0.067)
Design 2, $\gamma_0 = 0$								
Population	1.011	1.000	1.000	1.000	1.029	1.074	1.27	1.34
$T = 1000$	0.986	0.991	1.014	1.001	1.018	1.065	1.19	1.26
	(0.007)	(0.002)	(0.007)	(0.002)	(0.010)	(0.018)	(0.032)	(0.039)
$T = 10000$	1.011	1.000	1.001	1.000	1.042	1.075	1.34	1.27
	(0.007)	(0.001)	(0.002)	(0.000)	(0.011)	(0.017)	(0.038)	(0.039)
Design 3, $\gamma_0 = -0.10$								
Population	1.042	1.016	1.250	1.002	1.032	1.085	1.62	1.90
$T = 1000$	1.005	1.016	1.138	0.998	1.024	1.107	1.52	1.63
	(0.012)	(0.008)	(0.035)	(0.003)	(0.012)	(0.022)	(0.066)	(0.076)
$T = 10000$	1.011	1.000	1.248	1.003	1.019	1.087	1.57	1.88
	(0.014)	(0.008)	(0.037)	(0.003)	(0.011)	(0.018)	(0.067)	(0.081)
Design 4, $\beta_0 = 0.95$								
Population	1.008	1.002	1.036	1.000	1.021	1.060	1.18	1.27
$T = 1000$	0.987	0.994	1.036	0.998	1.022	1.039	1.19	1.23
	(0.006)	(0.003)	(0.010)	(0.002)	(0.010)	(0.016)	(0.029)	(0.034)
$T = 10000$	0.998	1.002	1.028	1.000	1.014	1.056	1.20	1.21
	(0.005)	(0.003)	(0.012)	(0.001)	(0.009)	(0.017)	(0.028)	(0.039)
Design 5, GED 1.5								
Population	1.022	1.007	1.130	1.000	1.022	1.053	1.51	1.71
$T = 1000$	1.013	1.003	1.072	1.001	1.009	1.014	1.41	1.55
	(0.010)	(0.005)	(0.020)	(0.002)	(0.014)	(0.020)	(0.051)	(0.062)
$T = 10000$	1.011	1.004	1.145	1.001	1.015	1.019	1.52	1.70
	(0.009)	(0.005)	(0.023)	(0.001)	(0.014)	(0.021)	(0.058)	(0.080)

Table 2: Variance ratios, sequential over J. Population values and Monte Carlo ratios over the converged samples of $R = 1000$, with bootstrap standard errors in parentheses.

Figure 1 plots the population ratio for ω against γ_0 for β_0 from 0.90 to 0.99. The ratio increases with $|\gamma_0|$ and with β_0, and each factor amplifies the effect of the other. At $\beta_0 = 0.90$, the ratio is at most 1.04, whereas at $\beta_0 = 0.99$ it reaches 1.70 at $\gamma_0 = -0.10$ and 2.93 at $\gamma_0 = -0.14$.

Overall, the simulation results confirm that the 2S estimator loses efficiency mainly in the mean log variance, through the estimation of ν, and that the 3S estimator recovers most of the efficiency for the mean parameters.

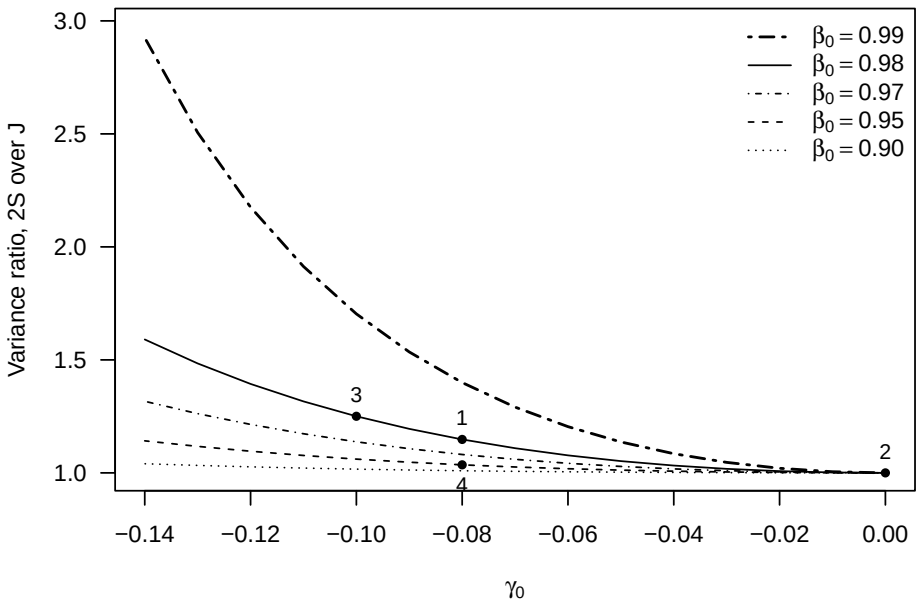


Figure 1: Population variance ratio of the 2S to the J estimator of ω against γ_0, with ν estimated and the other settings of Design 1, for $\beta_0 \in \{0.90, 0.95, 0.97, 0.98, 0.99\}$. Dots mark Designs 1 to 4.

7 Application to S&P 500 returns

The data are daily percentage log returns of the S&P 500 closing prices from January 1970 to August 2026, $T = 14{,}287$. We fit the AR(1)-EGARCH(1,1) model with $c = \sqrt{2/\pi}$, the value of $\mathsf{E}|\eta_t|$ under the Gaussian distribution. For the population ratios under a distribution L, we set $c = \mathsf{E}_L|\eta_t|$, the centering of Sections 2 to 6, and shift ω by $\alpha\{\mathsf{E}_L|\eta_t| - \sqrt{2/\pi}\}$ so that the filter remains unchanged. The ratios for ω therefore depend on this centering, whereas those for the mean log variance $\omega/(1-\beta)$ do not. Table 3 reports descriptive statistics.

	Mean	Standard deviation	Skewness	Raw kurtosis
Returns y_t	0.031	1.084	−0.94	26.50
Standardized J residuals	0.003	1.000	−0.49	5.86

Table 3: Descriptive statistics, S&P 500, January 1970 to August 2026.

Table 4 reports the estimates and their standard errors. Table 5 reports the population ratios evaluated at the J estimate under the Gaussian distribution and under a GED fitted by maximum likelihood to the standardized J residuals. The fitted GED has shape parameter 1.39 and fourth moment $\kappa_4 = 4.04$. At the J estimate, $\mathsf{E}[\log \Lambda_s^{(20)}(\hat{\boldsymbol{\theta}}_{\mathrm{J}})]$ equals -0.44 under the Gaussian distribution and -0.45 under the fitted GED, so $\mathrm{INV}(\hat{\boldsymbol{\theta}}_{\mathrm{J}})$ holds at this estimate.

With ν estimated and leverage present at $\hat{\gamma} = -0.10$, the 2S estimator is less efficient for ω, with variance ratios of 1.21 under the Gaussian distribution and 1.13 under the GED, and for γ and β, with ratios of 1.015 to 1.033, but not for α. The loss is concentrated in the mean log variance $\omega/(1-\beta)$, with ratios of 1.21 and 1.17.

	ν	ϕ	ω	α	γ	β
OLS	0.0309	−0.0187				
s.e.	0.0089	0.0198				
2S	0.0309	−0.0187	0.0004	0.1478	−0.0952	0.9770
s.e.	0.0089	0.0198	0.0020	0.0150	0.0103	0.0028
3S	0.0227	0.0478	0.0004	0.1478	−0.0952	0.9770
s.e.	0.0071	0.0091	0.0020	0.0150	0.0103	0.0028
J	0.0237	0.0446	0.0006	0.1459	−0.1005	0.9770
s.e.	0.0071	0.0090	0.0018	0.0146	0.0104	0.0028

Table 4: Estimates and standard errors, S&P 500. Standard errors are sample sandwich estimates with $\mathcal{J}$ estimated by $\frac{1}{2}\hat{C} + \text{diag}(\hat{W}, 0)$, heteroskedasticity-consistent for OLS, and include the first-step correction of (4.6) for 2S. The 3S estimator inherits the 2S variance estimates.

	2S				OLS		3S	
Distribution	ω	α	γ	β	ν	ϕ	ν	ϕ
Gaussian	1.208	1.000	1.033	1.019	1.51	1.74	1.029	1.079
GED 1.39	1.131	1.000	1.024	1.015	1.54	1.80	1.017	1.041

Table 5: Population variance ratios at the J estimate, sequential over J, with ν estimated and $c = \mathsf{E}|\eta_t|$ under each distribution.

Under the Gaussian distribution and at the same parameters with $\gamma_0 = 0$, the ratio would be 1.009 for γ and one for the other variance parameters, so the losses in ω and β arise mainly from estimating ν under leverage (Remark 5.2). With ν known, the ratio for ω at the J estimate is 1.0015.

For the mean parameters, OLS has 1.5 to 1.8 times the variance of the J estimator, while the 3S estimator is within 8% of it, so the J estimator is also the most efficient under both distributions. The sample standard errors in Table 4 imply 3S to J variance ratios of 1.00 for ν and 1.03 for ϕ, with 95% intervals from the block bootstrap of Künsch (1989) of 0.95 to 1.05 and 0.95 to 1.13, so the data do not discriminate between the two estimators for the mean parameters.

The OLS estimate of ϕ, −0.019, differs from the 3S and J estimates, 0.048 and 0.045, although all three are consistent for ϕ_0 under the model. With the covariance of each pair estimated from their influence functions, the specification tests of Hausman (1978) yield z-statistics of −3.6 for OLS against J, −4.0 for OLS against 3S and 0.9 for 3S against J. The tests therefore reject the AR(1) conditional mean. The rejection is consistent with the finding of LeBaron (1992) and Sentana and Wadhwani (1992) that first-order autocorrelations of daily index returns are larger in periods of low volatility and smaller in periods of high volatility. Both studies capture this pattern with an autoregressive coefficient that depends on the conditional variance, and a conditional mean that switches between volatility regimes is an alternative. Hence, the ratios in Table 5 describe the fitted model rather than the efficiency of the estimators in these data.

With innovations resampled from the standardized J residuals (Table 3) and sandwich covariances that include the cross terms in $\mathsf{E}[\eta_t^3]$, which vanish under (A1) (Remark 4.1), the 2S ratios

become 1.14 for ω and 0.99 for β, and the 3S estimator improves on the J estimator for ϕ, with ratio 0.95. Under the symmetrized empirical distribution of the residuals, the ratio is 0.97, close to the value 0.98 implied by Corollary 5.5 for $\kappa_4 = 5.86$ at $\gamma_0 = 0$, so skewness accounts for the remainder. The ranking of the variance parameters is unchanged, with the largest loss in ω and none in α. The sample OLS to J variance ratio for ϕ, 4.8 with a bootstrap 95% interval of 1.4 to 9.2, exceeds the population values of 1.7 and 1.8, consistent with a residual kurtosis above the values 3 and 4.04 of the two distributions.

8 Conclusion

Sequential estimation of an autoregressive mean and an EGARCH variance is common in applications, but the block diagonality of the information matrix that justifies it for GARCH errors fails for EGARCH. Without leverage, the first-step error enters the variance parameters only through $C_{\nu\gamma}$, so the 2S estimator loses only in γ and only when ν is estimated, including the case of demeaned returns. With leverage, the loss extends to ω and β at order γ_0^2 and increases with persistence, reaching approximately 15% for the mean log variance in the base design of the Monte Carlo study, and 17% to 21% at the S&P 500 estimate. These losses correspond to stopping after the second step. The 3S estimator attains the covariance W^{-1} of the infeasible GLS estimator, and without leverage the J estimator improves on it for ϕ if and only if $\kappa_4 < 5 + \tau$.

The analysis assumes symmetric innovations with exponential moments and the high-level expansions of (A5). Two extensions remain for future research. The first is an asymptotic theory that derives (A5) from INV($\boldsymbol{\theta}_0$) and primitive conditions when the filter has a kink in the mean parameters. The second extends the comparison to skewed innovations, where the cross terms of the two-step covariance no longer vanish, and to innovations without exponential moments, which (A2) excludes.

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A Supplementary Materials – Proofs

Throughout the Supplementary Materials, $|\cdot|$ is the Euclidean norm, $\|X\|_q = (\mathsf{E}|X|^q)^{1/q}$, $\partial_{\boldsymbol{\theta}} = \partial/\partial\boldsymbol{\theta}$, K denotes a finite constant whose value may change from line to line, and statements about random variables hold a.s. unless stated otherwise. Every sequence $X_t = f(\eta_t, \eta_{t-1}, \dots)$ with f measurable is strictly stationary and ergodic (Straumann and Mikosch, 2006, Proposition 2.5). We write

$$g(x) = \alpha_0(|x| - \mathsf{E}|\eta|) + \gamma_0 x, \qquad G(x) = \alpha_0|x| + \gamma_0 x, \tag{A.1}$$

so that

$$h_t = \omega_0 + g(\eta_{t-1}) + \beta_0 h_{t-1}, \qquad A_t = \beta_0 - \frac{1}{2}G(\eta_t). \tag{A.2}$$

Let $\Phi_{\boldsymbol{\phi}}(z) = 1 - \sum_i \phi_i z^i$, so that $\Phi(1) = \Phi_{\boldsymbol{\phi}_0}(1)$, and let ψ_k be the coefficients of $\Phi_{\boldsymbol{\phi}_0}(L)^{-1} = \sum_{k\geq 0}\psi_k L^k$. Finally, let

$$P_{t,j} = \prod_{l=1}^{j-1} A_{t-l} \qquad (P_{t,1} = 1). \tag{A.3}$$

Lemma A.1 (Stationarity and moments)**.** *Under (A1) to (A3), (2.1) has a unique strictly stationary solution, which is ergodic and given a.s. by*

$$h_t = \frac{\omega_0}{1-\beta_0} + \sum_{j\geq 1}\beta_0^{j-1} g(\eta_{t-j}), \qquad y_t - \nu_0 = \sum_{k\geq 0}\psi_k \varepsilon_{t-k}. \tag{A.4}$$

Moreover, $h_t \geq \underline{h}(\boldsymbol{\theta}_0)$, the variables σ_t and y_t have moments of all orders, and $\mathsf{E}[\sigma_t^{2\chi}]$ is bounded uniformly in $\gamma_0 \in [-\alpha_0, \alpha_0]$ for every $\chi > 0$.

***Proof of Lemma** A.1.* The existence and uniqueness of h_t follow from Theorem 2.8 of Straumann and Mikosch (2006), applied to the contraction $h \mapsto \omega_0 + g(\eta_{t-1}) + \beta_0 h$. By independence, $e^{wx} \leq 1 + w(e^x - 1)$ for $x \geq 0$ and $w \in [0,1]$, $1 + x \leq e^x$ and $0 \leq G(x) \leq 2\alpha_0|x|$,

$$\mathsf{E}[\sigma_t^{2\chi}] = e^{\chi \underline{h}(\boldsymbol{\theta}_0)} \prod_{j \geq 1} \mathsf{E}\big[e^{\chi \beta_0^{j-1} G(\eta)}\big] \leq \exp\Big\{\chi \underline{h}(\boldsymbol{\theta}_0) + \frac{\mathsf{E}[e^{2\chi\alpha_0|\eta|}] - 1}{1 - \beta_0}\Big\}, \tag{A.5}$$

which is finite under (A2) and does not depend on γ_0. Hence, $\varepsilon_t = \sigma_t \eta_t$ has moments of all orders, the series for y_t converges absolutely a.s. and in every L^r by Minkowski's inequality, and y_t is the unique stationary solution of the autoregression by the same theorem under the root condition in (A3). Both solutions are ergodic as functions of $(\eta_t, \eta_{t-1}, \dots)$. □

Let

$$\begin{aligned} \boldsymbol{w}_s &= \boldsymbol{d}_s + (\alpha_0 \operatorname{sgn} \eta_s + \gamma_0)\boldsymbol{b}_s, \\ \boldsymbol{d}_s &= [\mathbf{0}_{p+1}^\mathsf{T}, 1, |\eta_s| - \mathsf{E}|\eta|, \eta_s, h_s]^\mathsf{T}, \\ \boldsymbol{b}_s &= \sigma_s^{-1} \partial_{\boldsymbol{\theta}} \varepsilon_s(\boldsymbol{m}_0) = -\sigma_s^{-1}\big[\Phi(1), y_{s-1} - \nu_0, \dots, y_{s-p} - \nu_0, \mathbf{0}_4^\mathsf{T}\big]^\mathsf{T}. \end{aligned} \tag{A.6}$$

Let $z_s(\boldsymbol{\theta}) = \varepsilon_s(\boldsymbol{m}) e^{-h_s(\boldsymbol{\theta})/2}$. Differentiating $h_{s+1}(\boldsymbol{\theta}) = \omega + \alpha(|z_s(\boldsymbol{\theta})| - c) + \gamma z_s(\boldsymbol{\theta}) + \beta h_s(\boldsymbol{\theta})$ at $\boldsymbol{\theta}_0$ yields the linear stochastic recurrence equation $\partial_{\boldsymbol{\theta}} h_{s+1} = \boldsymbol{w}_s + A_s \partial_{\boldsymbol{\theta}} h_s$. Iterating it backward yields

$$\boldsymbol{a}_t = \sum_{j \geq 1} P_{t,j} \boldsymbol{w}_{t-j}. \tag{A.7}$$

Lemma A.2 (Differentiated recursion)**.** *Under (A1) to (A3), $\mathsf{E}[|\boldsymbol{w}_s|^2] < \infty$, the series (A.7) converges in L^2, and $\boldsymbol{a}_t$ is $\mathcal{F}_{t-1}$-measurable and satisfies $\boldsymbol{a}_t = \boldsymbol{w}_{t-1} + A_{t-1}\boldsymbol{a}_{t-1}$. Under (A4), $\boldsymbol{a}_t = \partial h_t(\boldsymbol{\theta}_0)/\partial \boldsymbol{\theta}$.*

***Proof of Lemma** A.2.* We first prove that $\boldsymbol{w}_s \in L^2$. The vector $\boldsymbol{d}_s$ is in L^2 because $\underline{h}(\boldsymbol{\theta}_0) \leq h_s \leq \sigma_s^2$ (Lemma A.1). The norm of $(\alpha_0 \operatorname{sgn} \eta_s + \gamma_0)\boldsymbol{b}_s$ is at most $2\alpha_0 e^{-\underline{h}(\boldsymbol{\theta}_0)/2}|\boldsymbol{\xi}_s|$, which is in L^2 because y_s has finite second moments.

We now prove convergence. Since $P_{t,j} \in \sigma(\eta_{t-j+1}, \dots, \eta_{t-1})$ is independent of $\boldsymbol{w}_{t-j} \in \mathcal{F}_{t-j}$ and the A_{t-l} are IID,

$$\|P_{t,j}\boldsymbol{w}_{t-j}\|_2 = \rho^{(j-1)/2}\|\boldsymbol{w}_0\|_2. \tag{A.8}$$

By (A.8), the series (A.7) converges in L^2, and $\boldsymbol{a}_t$ is a measurable function of $(\eta_{t-1}, \eta_{t-2}, \dots)$, hence $\mathcal{F}_{t-1}$-measurable. Multiplying the series for $\boldsymbol{a}_{t-1}$ by A_{t-1}, which is independent of it, yields the recursion.

Under (A4), $h_t(\boldsymbol{\theta}_0) = h_t$ by uniqueness and $\varepsilon_t \neq 0$ a.s., so (2.4) is differentiable at $(\boldsymbol{\theta}_0, h_t)$, and differentiating (2.2) yields $\boldsymbol{x}_{t+1} = \boldsymbol{w}_t + A_t \boldsymbol{x}_t$ for $\boldsymbol{x}_t = \partial h_t(\boldsymbol{\theta}_0)/\partial \boldsymbol{\theta}$. Since $\boldsymbol{x}_t$ is stationary, $\mathcal{F}_{t-1}$-measurable and bounded by $L_t \in L^2$, iterating yields $\|\boldsymbol{x}_t - \sum_{j \leq n} P_{t,j}\boldsymbol{w}_{t-j}\|_2 = \rho^{n/2}\|\boldsymbol{x}_0\|_2 \to 0$, so that $\boldsymbol{x}_t = \boldsymbol{a}_t$. □

Differentiating the Gaussian log quasi-likelihood l_t (at a parameter where the derivative of the filter exists) yields

$$\boldsymbol{s}_t(\boldsymbol{\theta}) = -\tfrac{1}{2}\{1 - z_t(\boldsymbol{\theta})^2\}\partial_{\boldsymbol{\theta}} h_t(\boldsymbol{\theta}) - z_t(\boldsymbol{\theta}) e^{-h_t(\boldsymbol{\theta})/2}\partial_{\boldsymbol{\theta}}\varepsilon_t(\boldsymbol{m}), \tag{A.9}$$

which recovers (3.1) with $\partial h_t/\partial\boldsymbol{\theta} = \boldsymbol{a}_t$ and $\sigma_t^{-1}\partial_{\boldsymbol{\theta}}\varepsilon_t = \boldsymbol{b}_t$ at $\boldsymbol{\theta}_0$.

***Proof of Proposition** 3.1.* Since $\mathsf{E}[(1-\eta_t^2)^2] = \kappa_4 - 1$, $\mathsf{E}[\eta_t^2] = 1$ and $\mathsf{E}[\eta_t(1-\eta_t^2)] = 0$ under (A1), conditioning on $\mathcal{F}_{t-1}$ yields $\mathcal{I}$ in (3.2).

We now prove that

$$\mathsf{E}[\boldsymbol{s}_t(\boldsymbol{\theta})] = -\big\{\tfrac{1}{2}C + \operatorname{diag}(W, 0)\big\}(\boldsymbol{\theta} - \boldsymbol{\theta}_0) + o(|\boldsymbol{\theta} - \boldsymbol{\theta}_0|), \tag{A.10}$$

which implies $\mathcal{J}$ in (3.2). Let $\underline{h}_{\mathcal{N}} = \min_{\mathcal{N}} \underline{h}$. Iterating (2.2) under the constraints of Θ yields $h_t(\boldsymbol{\theta}) \geq \underline{h}(\boldsymbol{\theta}) \geq \underline{h}_{\mathcal{N}}$ on $\mathcal{N}$. Let

$$\begin{aligned} \boldsymbol{\xi}_t(\boldsymbol{m}) &= -\big[\Phi_{\boldsymbol{\phi}}(1), y_{t-1} - \nu_0, \ldots, y_{t-p} - \nu_0\big]^{\mathsf{T}}, \\ \delta_t(\boldsymbol{m}) &= \varepsilon_t(\boldsymbol{m}) - \varepsilon_t = \boldsymbol{\xi}_t(\boldsymbol{m})^{\mathsf{T}}(\boldsymbol{m} - \boldsymbol{m}_0), \\ \upsilon_t(\boldsymbol{\theta}) &= 1 - \{\sigma_t^2 + \delta_t(\boldsymbol{m})^2\} e^{-h_t(\boldsymbol{\theta})}. \end{aligned} \tag{A.11}$$

Since $\mathsf{E}[\varepsilon_t(\boldsymbol{m}) \mid \mathcal{F}_{t-1}] = \delta_t(\boldsymbol{m})$ and $\mathsf{E}[\varepsilon_t(\boldsymbol{m})^2 \mid \mathcal{F}_{t-1}] = \sigma_t^2 + \delta_t(\boldsymbol{m})^2$,

$$\mathsf{E}[\boldsymbol{s}_t(\boldsymbol{\theta})] = -\tfrac{1}{2}\mathsf{E}\big[\upsilon_t(\boldsymbol{\theta})\partial_{\boldsymbol{\theta}} h_t(\boldsymbol{\theta})\big] - \mathsf{E}\big[\delta_t(\boldsymbol{m}) e^{-h_t(\boldsymbol{\theta})}\partial_{\boldsymbol{\theta}}\varepsilon_t(\boldsymbol{m})\big] =: -\tfrac{1}{2}\boldsymbol{q}_1(\boldsymbol{\theta}) - \boldsymbol{q}_2(\boldsymbol{\theta}). \tag{A.12}$$

Both expectations are finite on $\mathcal{N}$ by (A4) and Lemma A.1.

Let $\boldsymbol{u} = \boldsymbol{\theta} - \boldsymbol{\theta}_0$ and $\Delta = h_t(\boldsymbol{\theta}) - h_t$. By (A4) and Fubini's theorem, $x \mapsto h_t(\boldsymbol{\theta}_0 + x\boldsymbol{u})$ is Lipschitz with derivative $\partial_{\boldsymbol{\theta}} h_t(\boldsymbol{\theta}_0 + x\boldsymbol{u})^{\mathsf{T}}\boldsymbol{u}$ for almost every $x \in [0,1]$, so

$$\|\Delta - \boldsymbol{a}_t^{\mathsf{T}}\boldsymbol{u}\|_2 \leq |\boldsymbol{u}| \sup_{x\in[0,1]} \|\partial_{\boldsymbol{\theta}} h_t(\boldsymbol{\theta}_0 + x\boldsymbol{u}) - \boldsymbol{a}_t\|_2 = o(|\boldsymbol{u}|). \tag{A.13}$$

Since $\sigma_t^2 e^{-h_t(\boldsymbol{\theta})} = e^{-\Delta}$,

$$\upsilon_t(\boldsymbol{\theta}) - \boldsymbol{a}_t^{\mathsf{T}}\boldsymbol{u} = (1 - e^{-\Delta} - \Delta) + (\Delta - \boldsymbol{a}_t^{\mathsf{T}}\boldsymbol{u}) - \delta_t(\boldsymbol{m})^2 e^{-h_t(\boldsymbol{\theta})}. \tag{A.14}$$

The first term $(1 - e^{-\Delta} - \Delta)$ divided by $|\boldsymbol{u}|$ tends to zero a.s. and is dominated by $L_t(1 + \sigma_t^2 e^{-\underline{h}_{\mathcal{N}}}) \in L^2$. The last term $\delta_t(\boldsymbol{m})^2 e^{-h_t(\boldsymbol{\theta})}$ is $O(|\boldsymbol{u}|^2)$ in L^2. Hence, $\|\upsilon_t(\boldsymbol{\theta}) - \boldsymbol{a}_t^{\mathsf{T}}\boldsymbol{u}\|_2 = o(|\boldsymbol{u}|)$, so $\|\upsilon_t(\boldsymbol{\theta})\|_2 = O(|\boldsymbol{u}|)$, and the Cauchy–Schwarz inequality and (A4) imply

$$\boldsymbol{q}_1(\boldsymbol{\theta}) = \mathsf{E}[\upsilon_t(\boldsymbol{\theta})\boldsymbol{a}_t] + \mathsf{E}\big[\upsilon_t(\boldsymbol{\theta})\{\partial_{\boldsymbol{\theta}} h_t(\boldsymbol{\theta}) - \boldsymbol{a}_t\}\big] = \mathsf{E}[\boldsymbol{a}_t\boldsymbol{a}_t^{\mathsf{T}}]\boldsymbol{u} + o(|\boldsymbol{u}|) = C\boldsymbol{u} + o(|\boldsymbol{u}|). \tag{A.15}$$

The term $\boldsymbol{q}_2$ equals

$$\boldsymbol{q}_2(\boldsymbol{\theta}) = \mathsf{E}\big[e^{-h_t(\boldsymbol{\theta})}\partial_{\boldsymbol{\theta}}\varepsilon_t(\boldsymbol{m})\,\boldsymbol{\xi}_t(\boldsymbol{m})^{\mathsf{T}}\big](\boldsymbol{m} - \boldsymbol{m}_0). \tag{A.16}$$

The matrix is continuous at $\boldsymbol{\theta}_0$, because its entries are polynomials in $\boldsymbol{m}$ with coefficients bounded

by $K\{1+\sum_i (y_{t-i}-\nu_0)^2\}$, and

$$\mathsf{E}\Big[\big|e^{-h_t(\boldsymbol{\theta})}-e^{-h_t}\big|\Big\{1+\sum_{i=1}^{p}(y_{t-i}-\nu_0)^2\Big\}\Big]\to 0 \tag{A.17}$$

by the Lipschitz bound in (A4) and the dominated convergence theorem. At $\boldsymbol{\theta}_0$, its $\boldsymbol{m}$ rows equal W and its $\boldsymbol{v}$ rows are zero. Hence, $\partial \boldsymbol{q}_2(\boldsymbol{\theta}_0)/\partial\boldsymbol{\theta}^{\mathsf{T}} = \operatorname{diag}(W,0)$, and the expansion follows from (A.12).

It remains to prove that W and $\mathcal{J}$ are positive definite. The matrix W is positive definite because $\Phi(1)>0$ and no nontrivial linear combination of $1, y_{t-1},\dots,y_{t-p}$ is zero a.s. For $\boldsymbol{x}=[\boldsymbol{x}_{\boldsymbol{m}}^{\mathsf{T}},\boldsymbol{x}_{\boldsymbol{v}}^{\mathsf{T}}]^{\mathsf{T}}$,

$$\boldsymbol{x}^{\mathsf{T}}\mathcal{J}\boldsymbol{x} = \tfrac{1}{2}\mathsf{E}[(\boldsymbol{a}_t^{\mathsf{T}}\boldsymbol{x})^2] + \boldsymbol{x}_{\boldsymbol{m}}^{\mathsf{T}} W \boldsymbol{x}_{\boldsymbol{m}}. \tag{A.18}$$

Suppose that $\boldsymbol{x}_{\boldsymbol{m}}=\mathbf{0}$ and $\boldsymbol{a}_t^{\mathsf{T}}\boldsymbol{x}=0$ for every t. The recursion of Lemma A.2 implies $\boldsymbol{w}_t^{\mathsf{T}}\boldsymbol{x}=\boldsymbol{d}_t^{\mathsf{T}}\boldsymbol{x}=0$, and

$$x_\omega + x_\alpha(|\eta_t|-\mathsf{E}|\eta|)+x_\gamma\eta_t+x_\beta h_t = 0. \tag{A.19}$$

By conditioning on $\mathcal{F}_{t-1}$ and (A1), $x_\alpha = x_\gamma = 0$, and since h_t is nondegenerate under (A3), $x_\omega = x_\beta = 0$, so $\boldsymbol{x}_{\boldsymbol{v}}=\mathbf{0}$. Hence, $\mathcal{J}$ is positive definite. □

Let R_s change the sign of η_s. A random variable is *even* (*odd*) in s if it is invariant (changes sign) under R_s. Under (A1), R_s preserves the distribution of $(\eta_r)_{r\in\mathbb{Z}}$, so an integrable variable that is odd in some s has mean zero.

***Proof of Theorem** 3.2.* With $\gamma_0=0$, g is even, so h_t, $\sigma_t^{\pm1}$, A_t and $P_{t,j}$ are even in every r. The coordinates of $\boldsymbol{w}_s$ have the following parities.

(a) The coordinates 1, $|\eta_s|-\mathsf{E}|\eta|$ and h_s are even in every r.

(b) The coordinate η_s is odd in s only.

(c) The ν coordinate $-\alpha_0\Phi(1)\operatorname{sgn}(\eta_s)\sigma_s^{-1}$ is odd in s only.

(d) The kth term of the ϕ_i coordinate, $-\alpha_0\operatorname{sgn}(\eta_s)\sigma_s^{-1}\psi_k\sigma_{s-i-k}\eta_{s-i-k}$, is odd in s and in $s-i-k$.

By the Cauchy–Schwarz inequality and (A.8),

$$\sum_{j,j'}\mathsf{E}|P_{t,j}P_{t,j'}w_{t-j}^{(\ell)}w_{t-j'}^{(\ell')}| \le \|\boldsymbol{w}_0\|_2^2\sum_{j,j'}\rho^{(j+j'-2)/2}<\infty, \tag{A.20}$$

also after expanding $y_{s-i}-\nu_0=\sum_k\psi_k\varepsilon_{s-i-k}$, since $\sum_k|\psi_k|<\infty$. Hence, every entry of C used in (i) and (ii), and by the dominated convergence theorem every entry of W, is a sum of expectations of products of $P_{t,j}$, σ_t^{-2} and the coordinates (a) to (d), and each expectation is zero when its product is odd in some r.

We first prove (i). The generic product in $C_{\phi_i v}$ is

$$P_{t,j}P_{t,j'}\{-\alpha_0\operatorname{sgn}(\eta_{t-j})\sigma_{t-j}^{-1}\psi_k\sigma_u\eta_u\}\,w_{t-j'}^{(v)}, \qquad u=t-j-i-k<t-j. \tag{A.21}$$

For $v \in \{\omega, \alpha, \beta\}$, this product is odd in u. For $v = \gamma$, it is odd in u unless $t - j' = u$, in which case it is odd in $t - j$. The only odd factor in the terms of $C_{\nu v}$ and $C_{\gamma v}$ for $v \in \{\omega, \alpha, \beta\}$ is $\operatorname{sgn}(\eta_{t-j})$ or η_{t-j}, at index $t - j$. Finally,

$$W_{\nu\phi_i} = \Phi(1) \sum_{k \geq 0} \psi_k \mathsf{E}\big[\sigma_t^{-2} \sigma_u \eta_u\big], \qquad u = t - i - k, \tag{A.22}$$

where $\sigma_t^{-2}\sigma_u\eta_u$ is odd in u, so that $C_{\nu\phi_i} = \tau W_{\nu\phi_i} = 0$ by (iii) below. This proves the block structure of C and W, and that of $\mathcal{I}$ and $\mathcal{J}$ follows from (3.2).

We now prove (ii). In

$$C_{\nu\gamma} = \sum_{j,j' \geq 1} \mathsf{E}\big[P_{t,j} P_{t,j'} \{-\alpha_0 \Phi(1) \operatorname{sgn}(\eta_{t-j}) \sigma_{t-j}^{-1}\} \eta_{t-j'}\big] \tag{A.23}$$

the products with $j \neq j'$ are odd in $t - j$. For $j = j'$, $P_{t,j}^2$, $|\eta_{t-j}|$ and σ_{t-j}^{-1} depend on disjoint sets of innovations, so the term equals $-\alpha_0 \Phi(1) \rho^{j-1} \mathsf{E}|\eta| \, \mathsf{E}[\sigma^{-1}]$ and summing over j yields (3.4). The same argument yields $C_{\gamma\gamma} = \sum_j \rho^{j-1} = 1/(1-\rho)$.

It remains to prove (iii). Since the mean coordinates of $\boldsymbol{w}_s$ are $\alpha_0 \operatorname{sgn}(\eta_s) \sigma_s^{-1} \boldsymbol{\xi}_s$,

$$C_{\boldsymbol{m}\boldsymbol{m}} = \alpha_0^2 \sum_{j,j' \geq 1} \mathsf{E}\big[P_{t,j} P_{t,j'} \operatorname{sgn}(\eta_{t-j}) \operatorname{sgn}(\eta_{t-j'}) \sigma_{t-j}^{-1} \sigma_{t-j'}^{-1} \boldsymbol{\xi}_{t-j} \boldsymbol{\xi}_{t-j'}^{\mathsf{T}}\big]. \tag{A.24}$$

For $j \neq j'$, let $m = \min(j, j')$. Then $\operatorname{sgn}(\eta_{t-m})$ is the only factor that is odd in $t - m$, since every other factor is even in η_{t-m} or does not involve it, so these terms vanish. For $j = j'$, $\operatorname{sgn}(\eta_{t-j})^2 = 1$, and $P_{t,j}^2$ is independent of $\sigma_{t-j}^{-2} \boldsymbol{\xi}_{t-j} \boldsymbol{\xi}_{t-j}^{\mathsf{T}}$, so

$$C_{\boldsymbol{m}\boldsymbol{m}} = \alpha_0^2 \sum_{j \geq 1} \rho^{j-1} \mathsf{E}\big[\sigma_t^{-2} \boldsymbol{\xi}_t \boldsymbol{\xi}_t^{\mathsf{T}}\big] = \tau W. \tag{A.25}$$

Finally, $W_{\nu\nu} = \Phi(1)^2 \mathsf{E}[\sigma_t^{-2}]$ and $C_{\nu\gamma}^2 / C_{\gamma\gamma} = \tau \varpi W_{\nu\nu}$, which implies the second identity in (iii). Under (A1), Jensen's inequality implies $\varpi \in (0, 1)$. □

Proof of Proposition 3.3. We first prove analyticity. Let $U \subset \mathbb{C}$ be an open disc and $F(\gamma)$ a random function that is analytic on U with $\sup_U \mathsf{E}|F(\gamma)| < \infty$. By Cauchy's integral formula and Fubini's theorem, $\gamma \mapsto \mathsf{E}[F(\gamma)]$ is analytic on U. Extend the model to complex γ by

$$\begin{aligned} G_\gamma(x) &= \alpha_0 |x| + \gamma x, \qquad & A_s(\gamma) &= \beta_0 - \tfrac{1}{2} G_\gamma(\eta_s), \\ h_s(\gamma) &= h_s(0) + \gamma \zeta_s, \qquad & \zeta_s &= \sum_{l \geq 1} \beta_0^{l-1} \eta_{s-l}. \end{aligned} \tag{A.26}$$

Let $\rho(0)$ be ρ at $\gamma_0 = 0$ and $\bar{\gamma}_1 := \min\{\alpha_0, \sqrt{1 - \rho(0)}\}$. Under the symmetry in (A1),

$$\mathsf{E}|A(\gamma)|^2 = \rho(0) + \tfrac{1}{4}|\gamma|^2 \leq \bar{\rho} := \rho(0) + \tfrac{1}{4}\{1 - \rho(0)\} < 1 \qquad (|\gamma| \leq \bar{\gamma}_1). \tag{A.27}$$

In particular, (A3) holds for real $|\gamma_0| < \bar{\gamma}_1$. For $|\gamma| \leq \bar{\gamma}_1$, since $|e^{\frac{1}{2}\{h_u(\gamma) - h_{t-j}(\gamma)\}}| =$

$\sigma_u(\mathrm{Re}\,\gamma)/\sigma_{t-j}(\mathrm{Re}\,\gamma)$, Lemma A.1 bounds each term of the expansion of $C_{\phi_i v}$ in the proof of Theorem 3.2 by $K|\psi_k|\bar{\rho}^{(j+j'-2)/2}$, uniformly in γ. The series is therefore a uniformly convergent sum of analytic functions, hence analytic. The other entries of C are treated in the same way.

We now prove the parity in γ_0. Let R change the sign of every η_r. Under (A1), R preserves the distribution, maps $h_s(\gamma_0)$ to $h_s(-\gamma_0)$ and maps $\boldsymbol{a}_t(\gamma_0)$ to $\Sigma\boldsymbol{a}_t(-\gamma_0)$, where Σ changes the sign of the ν and γ coordinates, so

$$C(\gamma_0) = \Sigma C(-\gamma_0)\Sigma. \tag{A.28}$$

Hence, $C_{\phi_i\gamma}$ is odd and $C_{\phi_i v}$ is even in γ_0 for $v \in \{\omega, \alpha, \beta\}$. Likewise, $C_{\nu v}$ and $C_{\gamma v}$ are odd. Being odd and analytic, $C_{\phi_i\gamma}$, $C_{\nu v}$ and $C_{\gamma v}$ are $O(\gamma_0)$. Being even, analytic and equal to zero at $\gamma_0 = 0$ by Theorem 3.2(i), $C_{\phi_i v}$ is $O(\gamma_0^2)$. □

Lemma A.3 (OLS step)**.** *Under (A1) to (A3), (4.1) holds with* $\mathsf{E}[\boldsymbol{p}_t\boldsymbol{p}_t^\mathsf{T}] = V_{\mathrm{OLS}}$.

Proof of Lemma A.3. The OLS coefficients $\hat{\boldsymbol{\pi}}$ of y_t on $\boldsymbol{x}_t = [1, y_{t-1}, \ldots, y_{t-p}]^\mathsf{T}$, with $\boldsymbol{\pi}_0 = [\nu_0\Phi(1), \boldsymbol{\phi}_0^\mathsf{T}]^\mathsf{T}$, satisfy

$$\sqrt{T}(\hat{\boldsymbol{\pi}} - \boldsymbol{\pi}_0) = \Big\{\frac{1}{T}\sum_t \boldsymbol{x}_t\boldsymbol{x}_t^\mathsf{T}\Big\}^{-1}\frac{1}{\sqrt{T}}\sum_t \boldsymbol{x}_t\varepsilon_t. \tag{A.29}$$

The last sum has second moment $\mathsf{E}[\varepsilon_t^2|\boldsymbol{x}_t|^2] < \infty$, so the ergodic theorem allows replacing the inverse by its limit $\mathsf{E}[\boldsymbol{x}_t\boldsymbol{x}_t^\mathsf{T}]^{-1}$ up to $o_p(1)$, the limit being nonsingular by the proof of Proposition 3.1. A mean-value expansion of $\nu = \pi_1/\Phi_{\boldsymbol{\phi}}(1)$ yields the stated expansion and the covariance V_{OLS}. □

Lemma A.4 (Joint central limit theorem)**.** *Under (A1) to (A3),*

$$T^{-1/2}\sum_{t=1}^{T}\boldsymbol{v}_t \xrightarrow{\mathrm{d}} \mathrm{N}\big(\boldsymbol{0}, \mathsf{E}[\boldsymbol{v}_t\boldsymbol{v}_t^\mathsf{T}]\big), \qquad \boldsymbol{v}_t = [\boldsymbol{p}_t^\mathsf{T}, \boldsymbol{s}_t^\mathsf{T}, \eta_t\boldsymbol{b}_t^{(\boldsymbol{m})\mathsf{T}}]^\mathsf{T}, \tag{A.30}$$

and $\mathsf{E}[\boldsymbol{p}_t\boldsymbol{s}_t^{(\boldsymbol{v})\mathsf{T}}] = \boldsymbol{0}$.

Proof of Lemma A.4. $\boldsymbol{v}_t$ is stationary, ergodic and square-integrable. Each coordinate is a linear combination of η_t and $1-\eta_t^2$ with $\mathcal{F}_{t-1}$-measurable coefficients, so $\boldsymbol{v}_t$ is a martingale difference sequence, and the central limit theorem of Billingsley (1961) and the Cramér–Wold theorem imply asymptotic normality. Since $\boldsymbol{p}_t = -M^{-1}\sigma_t\boldsymbol{\xi}_t\eta_t$ and $\boldsymbol{s}_t^{(\boldsymbol{v})} = -\frac{1}{2}(1-\eta_t^2)\boldsymbol{a}_t^{(\boldsymbol{v})}$, conditioning on $\mathcal{F}_{t-1}$ and the symmetry in (A1) yield

$$\mathsf{E}\big[\boldsymbol{p}_t\boldsymbol{s}_t^{(\boldsymbol{v})\mathsf{T}}\big] = \tfrac{1}{2}M^{-1}\mathsf{E}\big[\sigma_t\boldsymbol{\xi}_t\boldsymbol{a}_t^{(\boldsymbol{v})\mathsf{T}}\big]\,\mathsf{E}\big[\eta_t(1-\eta_t^2)\big] = \boldsymbol{0}. \tag{A.31}$$

□

Proof of Proposition 4.1. By (A5), each estimator is asymptotically linear in the vector of Lemma A.4, so the Slutsky theorem implies asymptotic normality. The covariance of the J estimator is $\mathcal{J}^{-1}\mathcal{I}\mathcal{J}^{-1}$. For the 2S estimator, by Lemmas A.3 and A.4,

$$\mathsf{E}\big[\big(\boldsymbol{s}_t^{(\boldsymbol{v})} - \mathcal{J}_{\boldsymbol{vm}}\boldsymbol{p}_t\big)\big(\boldsymbol{s}_t^{(\boldsymbol{v})} - \mathcal{J}_{\boldsymbol{vm}}\boldsymbol{p}_t\big)^\mathsf{T}\big] = \mathcal{I}_{\boldsymbol{vv}} + \mathcal{J}_{\boldsymbol{vm}}V_{\mathrm{OLS}}\mathcal{J}_{\boldsymbol{mv}}. \tag{A.32}$$

For the 3S estimator, $\mathsf{E}[\eta_t^2 \boldsymbol{b}_t^{(\boldsymbol{m})} \boldsymbol{b}_t^{(\boldsymbol{m})\mathsf{T}}] = W$, so the covariance is $W^{-1}WW^{-1} = W^{-1}$. □

***Proof of Theorem** 5.1.* By (4.4) and (4.3), the two estimators of $\boldsymbol{v}$ have influence functions

$$\mathcal{J}_{\boldsymbol{vv}}^{-1}\big(\boldsymbol{s}_t^{(\boldsymbol{v})} - \mathcal{J}_{\boldsymbol{vm}}\boldsymbol{\iota}_t\big), \qquad \boldsymbol{\iota}_t = \boldsymbol{p}_t \text{ (2S)}, \qquad \boldsymbol{\iota}_t = (\mathcal{J}^{-1}\boldsymbol{s}_t)^{(\boldsymbol{m})} \text{ (J)}. \tag{A.33}$$

Hence, for $a \in \{2\mathrm{S}, \mathrm{J}\}$,

$$V_{a,\boldsymbol{vv}} = \mathcal{J}_{\boldsymbol{vv}}^{-1}\Big\{\mathcal{I}_{\boldsymbol{vv}} + \mathcal{J}_{\boldsymbol{vm}}\mathsf{E}[\boldsymbol{\iota}_t\boldsymbol{\iota}_t^\mathsf{T}]\mathcal{J}_{\boldsymbol{mv}} - \mathcal{J}_{\boldsymbol{vm}}\mathsf{E}\big[\boldsymbol{\iota}_t\boldsymbol{s}_t^{(\boldsymbol{v})\mathsf{T}}\big] - \mathsf{E}\big[\boldsymbol{s}_t^{(\boldsymbol{v})}\boldsymbol{\iota}_t^\mathsf{T}\big]\mathcal{J}_{\boldsymbol{mv}}\Big\}\mathcal{J}_{\boldsymbol{vv}}^{-1}. \tag{A.34}$$

For the 2S estimator, $\mathsf{E}[\boldsymbol{p}_t\boldsymbol{p}_t^\mathsf{T}] = V_{\mathrm{OLS}}$ and $\mathsf{E}[\boldsymbol{p}_t\boldsymbol{s}_t^{(\boldsymbol{v})\mathsf{T}}] = \mathbf{0}$ by Lemmas A.3 and A.4, so that

$$V_{2\mathrm{S},\boldsymbol{vv}} = \mathcal{J}_{\boldsymbol{vv}}^{-1}\big(\mathcal{I}_{\boldsymbol{vv}} + \mathcal{J}_{\boldsymbol{vm}}V_{\mathrm{OLS}}\mathcal{J}_{\boldsymbol{mv}}\big)\mathcal{J}_{\boldsymbol{vv}}^{-1}. \tag{A.35}$$

For the J estimator, $\mathsf{E}[\boldsymbol{\iota}_t\boldsymbol{\iota}_t^\mathsf{T}] = V_{\mathrm{J},\boldsymbol{mm}}$ and $\mathsf{E}[\boldsymbol{\iota}_t\boldsymbol{s}_t^{(\boldsymbol{v})\mathsf{T}}] = (\mathcal{J}^{-1}\mathcal{I})_{\boldsymbol{mv}}$. By (3.2), $\mathcal{I}_{\cdot\boldsymbol{v}} = \frac{\kappa_4-1}{2}\mathcal{J}_{\cdot\boldsymbol{v}}$, so $(\mathcal{J}^{-1}\mathcal{I})_{\boldsymbol{mv}} = \frac{\kappa_4-1}{2}(\mathcal{J}^{-1}\mathcal{J})_{\boldsymbol{mv}} = 0$ and

$$V_{\mathrm{J},\boldsymbol{vv}} = \mathcal{J}_{\boldsymbol{vv}}^{-1}\big(\mathcal{I}_{\boldsymbol{vv}} + \mathcal{J}_{\boldsymbol{vm}}V_{\mathrm{J},\boldsymbol{mm}}\mathcal{J}_{\boldsymbol{mv}}\big)\mathcal{J}_{\boldsymbol{vv}}^{-1}. \tag{A.36}$$

Subtracting the two covariances yields (5.1). □

***Proof of Corollary** 5.2.* By Theorem 3.2, $\mathcal{J}$ and $\mathcal{I}$ are block diagonal, and $\mathcal{J}_{\boldsymbol{vm}}$ has the single nonzero entry $\mathcal{J}_{\gamma\nu} = \frac{1}{2}C_{\nu\gamma}$ by (3.2). By Theorem 5.1, the difference is therefore zero except in the (γ,γ) entry, which equals

$$\Big(\frac{\mathcal{J}_{\gamma\nu}}{\mathcal{J}_{\gamma\gamma}}\Big)^2\big(V_{\mathrm{OLS},\nu\nu} - V_{\mathrm{J},\nu\nu}\big) = \Big(\frac{C_{\nu\gamma}}{C_{\gamma\gamma}}\Big)^2\big(V_{\mathrm{OLS},\nu\nu} - V_{\mathrm{J},\nu\nu}\big). \tag{A.37}$$

Let $C_{\nu\nu\cdot\gamma} = C_{\nu\nu} - C_{\nu\gamma}^2/C_{\gamma\gamma}$, which is $\tau(1-\varpi)W_{\nu\nu}$ by Theorem 3.2(iii). Inverting the 2×2 block of $\mathcal{J}$ for (ν,γ) in $V_{\mathrm{J}} = \mathcal{J}^{-1}\mathcal{I}\mathcal{J}^{-1}$, with (3.2) for $\mathcal{I}$, yields

$$V_{\mathrm{J},\nu\nu} = \frac{\frac{\kappa_4-1}{4}C_{\nu\nu\cdot\gamma} + W_{\nu\nu}}{\big(\frac{1}{2}C_{\nu\nu\cdot\gamma} + W_{\nu\nu}\big)^2} = \frac{\mathcal{R}\big(\kappa_4, \tau(1-\varpi)\big)}{W_{\nu\nu}}. \tag{A.38}$$

Positivity of the difference follows from $V_{\mathrm{OLS},\nu\nu} \geq (W^{-1})_{\nu\nu} = W_{\nu\nu}^{-1}$ (Proposition 5.4 and Theorem 3.2(i)) and $\mathcal{R}(\kappa_4, x) < 1$ for $\kappa_4 < 5 + x$ by (5.2). With ν known, $\mathcal{J}_{\boldsymbol{vm}} = \frac{1}{2}C_{\boldsymbol{v}\boldsymbol{\phi}} = 0$ and (5.1) vanishes. □

***Proof of Corollary** 5.3.* With ν known, Theorem 5.1 implies

$$V_{2\mathrm{S},\boldsymbol{vv}} - V_{\mathrm{J},\boldsymbol{vv}} = \mathsf{B}\,G\,\mathsf{B}^\mathsf{T}, \qquad \mathsf{B} = C_{\boldsymbol{vv}}^{-1}C_{\boldsymbol{v}\boldsymbol{\phi}}, \qquad G = V_{\mathrm{OLS}} - V_{\mathrm{J},\boldsymbol{mm}}, \tag{A.39}$$

all evaluated at γ_0. By (3.6) and the positive definiteness of $C_{\boldsymbol{vv}}(0) = \mathrm{diag}(C_{\gamma\gamma}, C_{\bar{\boldsymbol{v}}\bar{\boldsymbol{v}}})(0)$, the γ row of B is $O(\gamma_0)$ and the $\bar{\boldsymbol{v}}$ rows are $O(\gamma_0^2)$. Since C is analytic, and the dominated convergence theorem under (A2) and Lemma A.1 implies that W, M and $\mathsf{E}[\sigma_t^2\boldsymbol{\xi}_t\boldsymbol{\xi}_t^\mathsf{T}]$ are continuous, G and V_{J}

are continuous at $\gamma_0 = 0$. Hence, G is bounded near $\gamma_0 = 0$, and for $a \in \{\gamma, \omega, \alpha, \beta\}$,

$$\left|(V_{2\mathrm{S},\boldsymbol{vv}} - V_{\mathrm{J},\boldsymbol{vv}})_{aa}\right| = \left|\mathsf{B}_{a\cdot} G\, \mathsf{B}_{a\cdot}^{\mathsf{T}}\right| \le |\mathsf{B}_{a\cdot}|^2 |G| = \begin{cases} O(\gamma_0^2), & a = \gamma, \\ O(\gamma_0^4), & a \neq \gamma. \end{cases} \tag{A.40}$$

By the proof of Proposition 3.1 with $\kappa_4 > 1$, $\mathcal{I}$ is positive definite, so $V_{\mathrm{J}} = \mathcal{J}^{-1}\mathcal{I}\mathcal{J}^{-1}$ is positive definite and $V_{\mathrm{J},aa}$ is bounded away from zero near $\gamma_0 = 0$. Dividing the bound by $V_{\mathrm{J},aa}$ yields (5.4). □

***Proof of Proposition** 5.4.* W is positive definite under (A1) to (A3) by the proof of Proposition 3.1. OLS and the infeasible GLS estimator are instrumental variables estimators with instruments $\boldsymbol{\xi}_t$ and $\sigma_t^{-2}\boldsymbol{\xi}_t$, based on the conditional moment restriction $\mathsf{E}[\varepsilon_t \mid \mathcal{F}_{t-1}] = 0$. Since $\sigma_t^{-2}\boldsymbol{\xi}_t$ are the efficient instruments for this restriction, Theorem 5.3 of Newey and McFadden (1994) implies that $V_{\mathrm{OLS}} - W^{-1}$ is positive semidefinite. □

***Proof of Corollary** 5.5.* By Theorem 3.2,

$$\mathcal{J}_{\boldsymbol{\phi\phi}} = \Big(1 + \frac{\tau}{2}\Big) W_{\boldsymbol{\phi\phi}}, \qquad \mathcal{I}_{\boldsymbol{\phi\phi}} = \Big\{1 + \frac{(\kappa_4 - 1)\tau}{4}\Big\} W_{\boldsymbol{\phi\phi}}, \tag{A.41}$$

and the off-diagonal blocks of the $\boldsymbol{\phi}$ rows vanish. Hence,

$$V_{\mathrm{J},\boldsymbol{\phi\phi}} = \mathcal{J}_{\boldsymbol{\phi\phi}}^{-1}\mathcal{I}_{\boldsymbol{\phi\phi}}\mathcal{J}_{\boldsymbol{\phi\phi}}^{-1} = \mathcal{R}(\kappa_4, \tau)\, W_{\boldsymbol{\phi\phi}}^{-1}. \tag{A.42}$$

Since the 3S covariance W^{-1} is block diagonal between ν and $\boldsymbol{\phi}$ by Theorem 3.2(i), and $V_{\mathrm{J},\nu\nu}$ is given in (5.3), the comparison reduces to $\mathcal{R}(\kappa_4, x) \le 1$ with $x = \tau$ and $x = \tau(1 - \varpi)$, which holds if and only if $\kappa_4 \le 5 + x$ by (5.2). □

B Variance ratios for α and β

	α			β		
	Population	$T = 1000$	$T = 10000$	Population	$T = 1000$	$T = 10000$
Design 1	1.000	0.993	1.001	1.012	1.017	1.003
		(0.003)	(0.001)		(0.007)	(0.007)
Design 2, $\gamma_0 = 0$	1.000	0.992	0.999	1.000	1.002	1.001
		(0.002)	(0.001)		(0.009)	(0.001)
Design 3, $\gamma_0 = -0.10$	1.000	1.001	0.999	1.029	1.030	1.018
		(0.005)	(0.001)		(0.012)	(0.010)
Design 4, $\beta_0 = 0.95$	1.000	0.990	0.997	1.001	1.002	0.998
		(0.003)	(0.001)		(0.004)	(0.002)
Design 5, GED 1.5	1.000	0.994	1.000	1.010	1.006	1.006
		(0.004)	(0.001)		(0.005)	(0.006)

Table B.1: Variance ratios of the 2S over the J estimator for α and β, with ν estimated. Population values and Monte Carlo ratios over the converged samples of $R = 1000$, with bootstrap standard errors in parentheses.